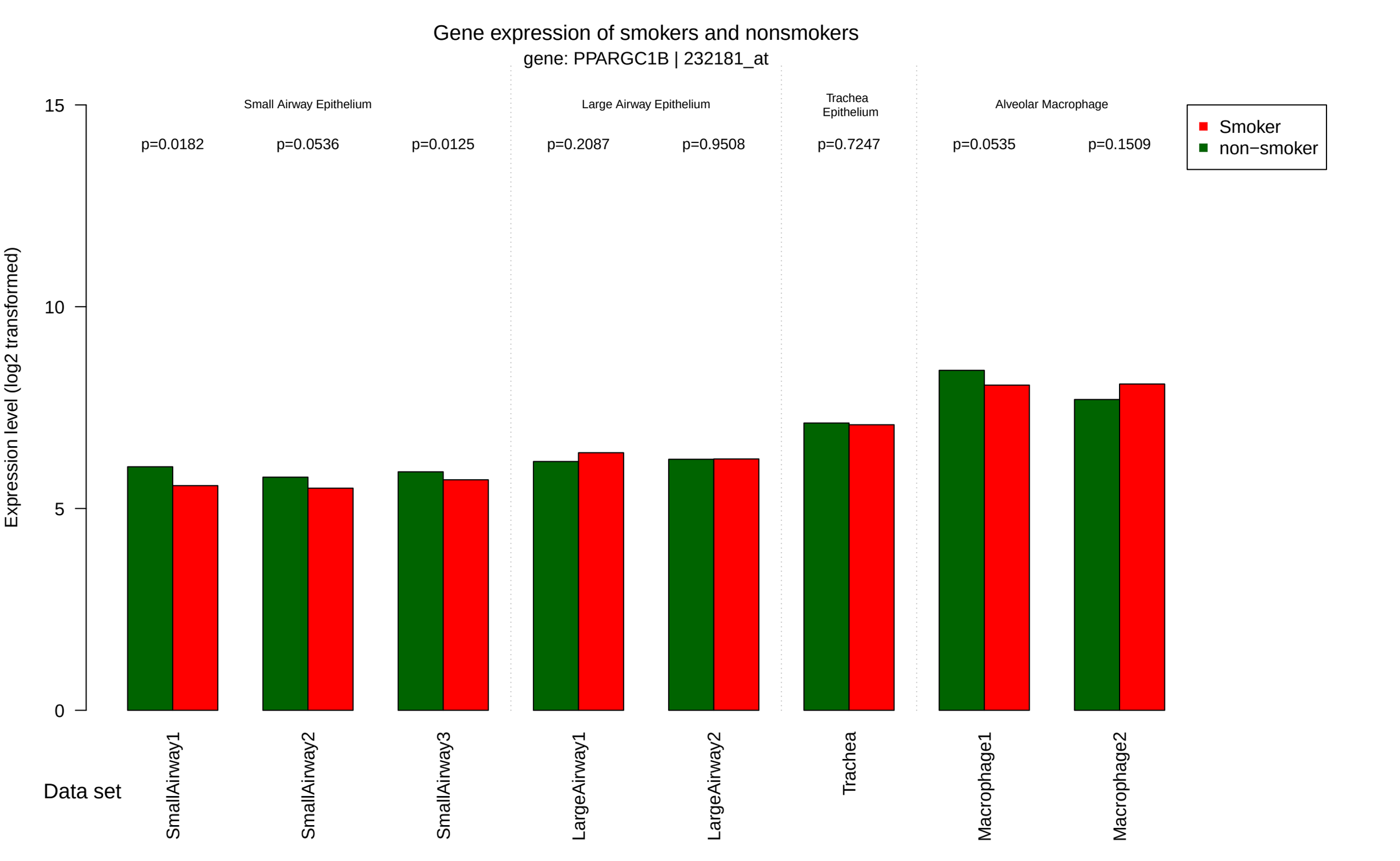
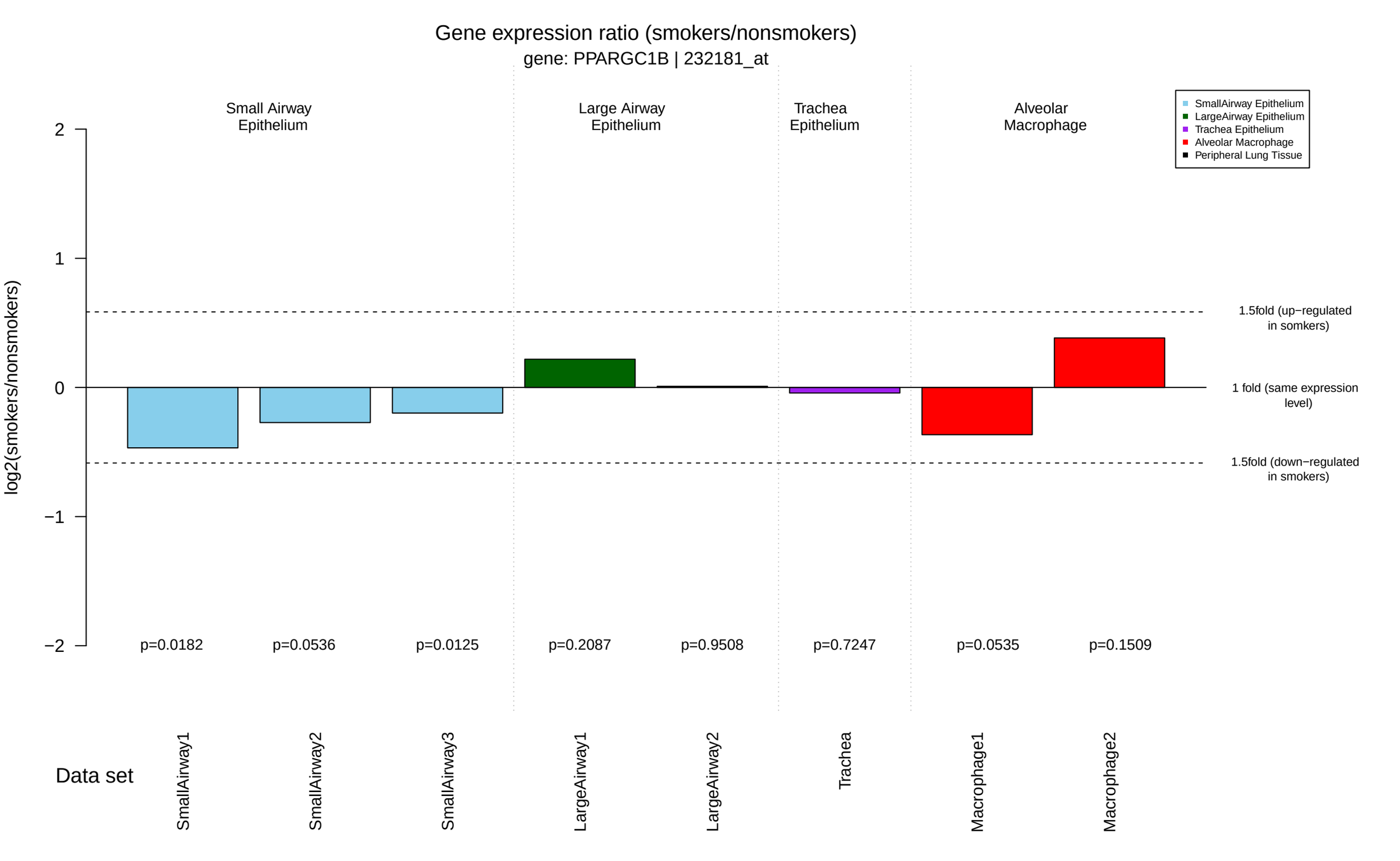
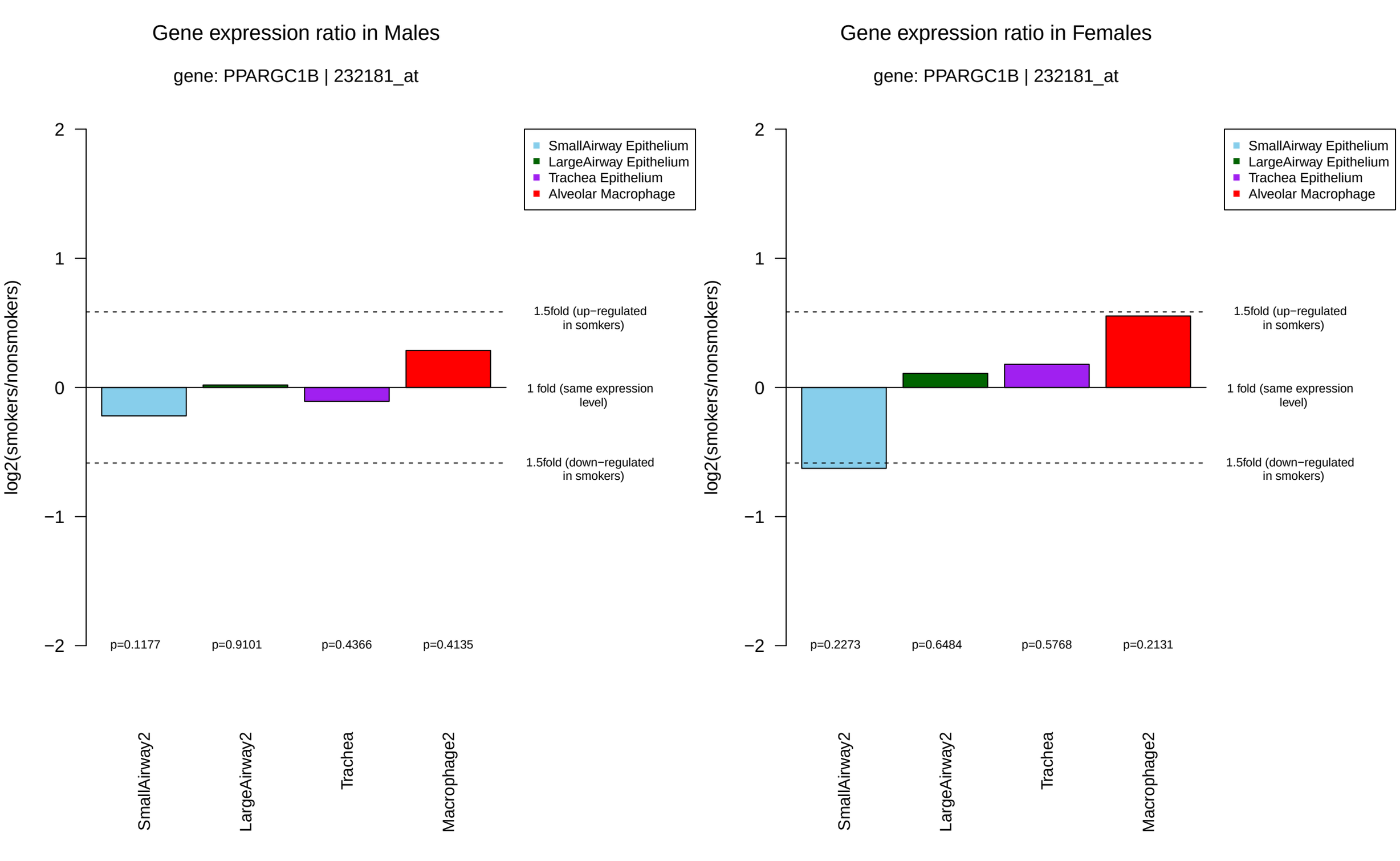
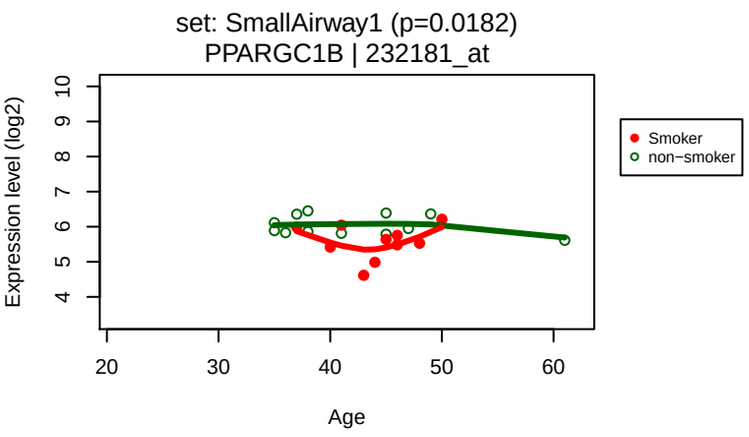
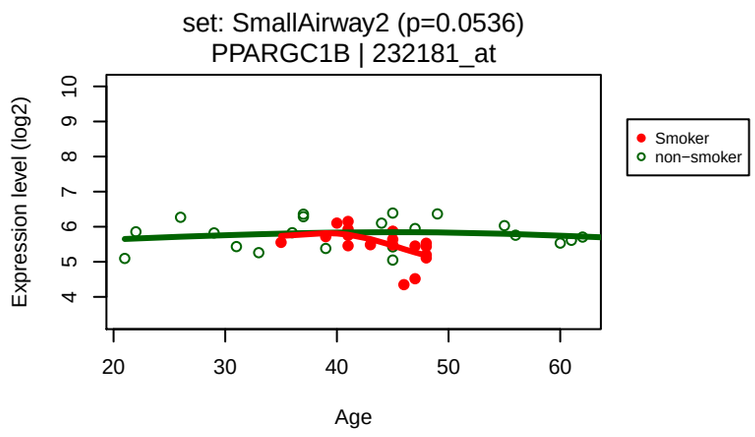
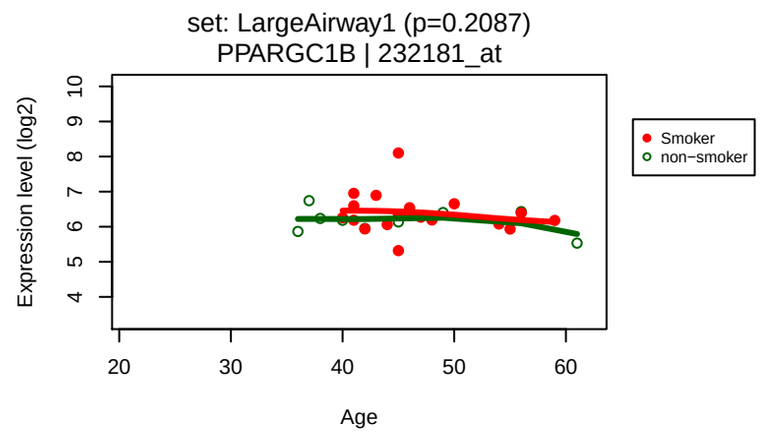

Gene expression of smokers and nonsmokers
gene: PPARGC1B | 232181_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0182
p=0.0536
p=0.0125
p=0.2087
p=0.9508
p=0.7247
p=0.0535
p=0.1509
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

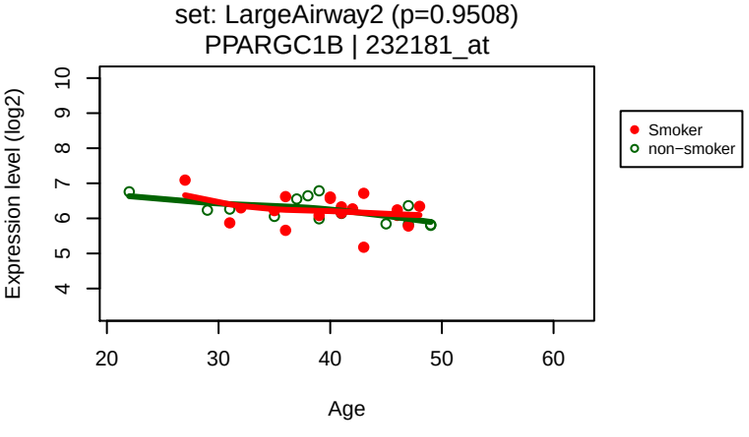

Gene expression ratio (smokers/nonsmokers)
gene: PPARGC1B | 232181_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0182
p=0.0536
p=0.0125
p=0.2087
p=0.9508
p=0.7247
p=0.0535
p=0.1509
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

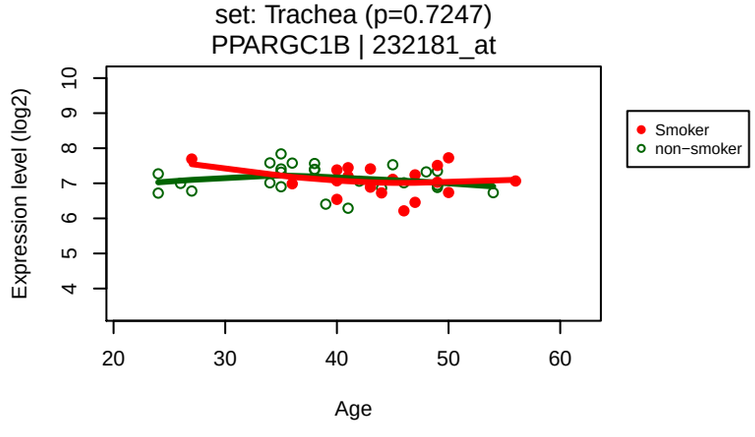

Gene expression ratio in Males
gene: PPARGC1B | 232181_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1177
p=0.9101
p=0.4366
p=0.4135
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPARGC1B | 232181_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2273
p=0.6484
p=0.5768
p=0.2131
SmallAirway2
LargeAirway2
Trachea
Macrophage2

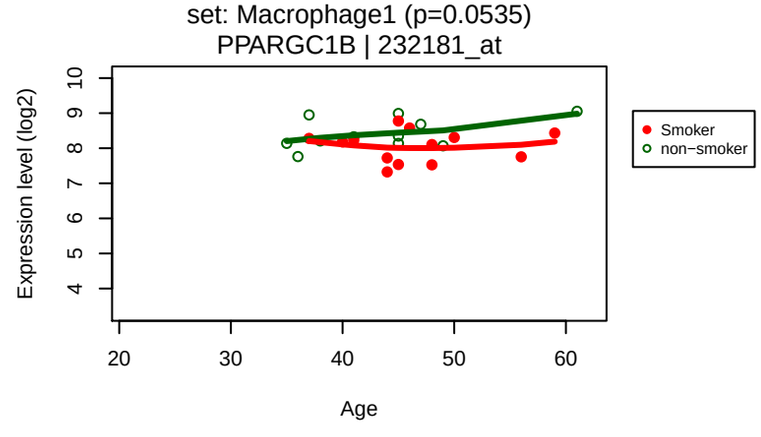

set: SmallAirway1 (p=0.0182)
PPARGC1B | 232181_at
Expression level (log2)
Age
Smoker
non−smoker

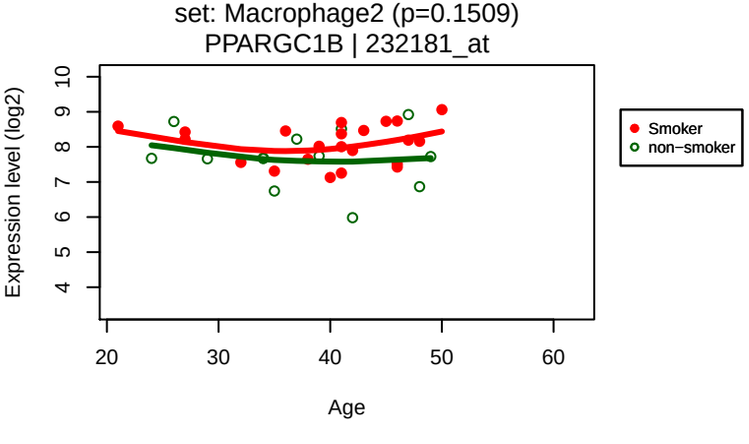

set: SmallAirway2 (p=0.0536)
PPARGC1B | 232181_at
Expression level (log2)
Age
Smoker
non−smoker

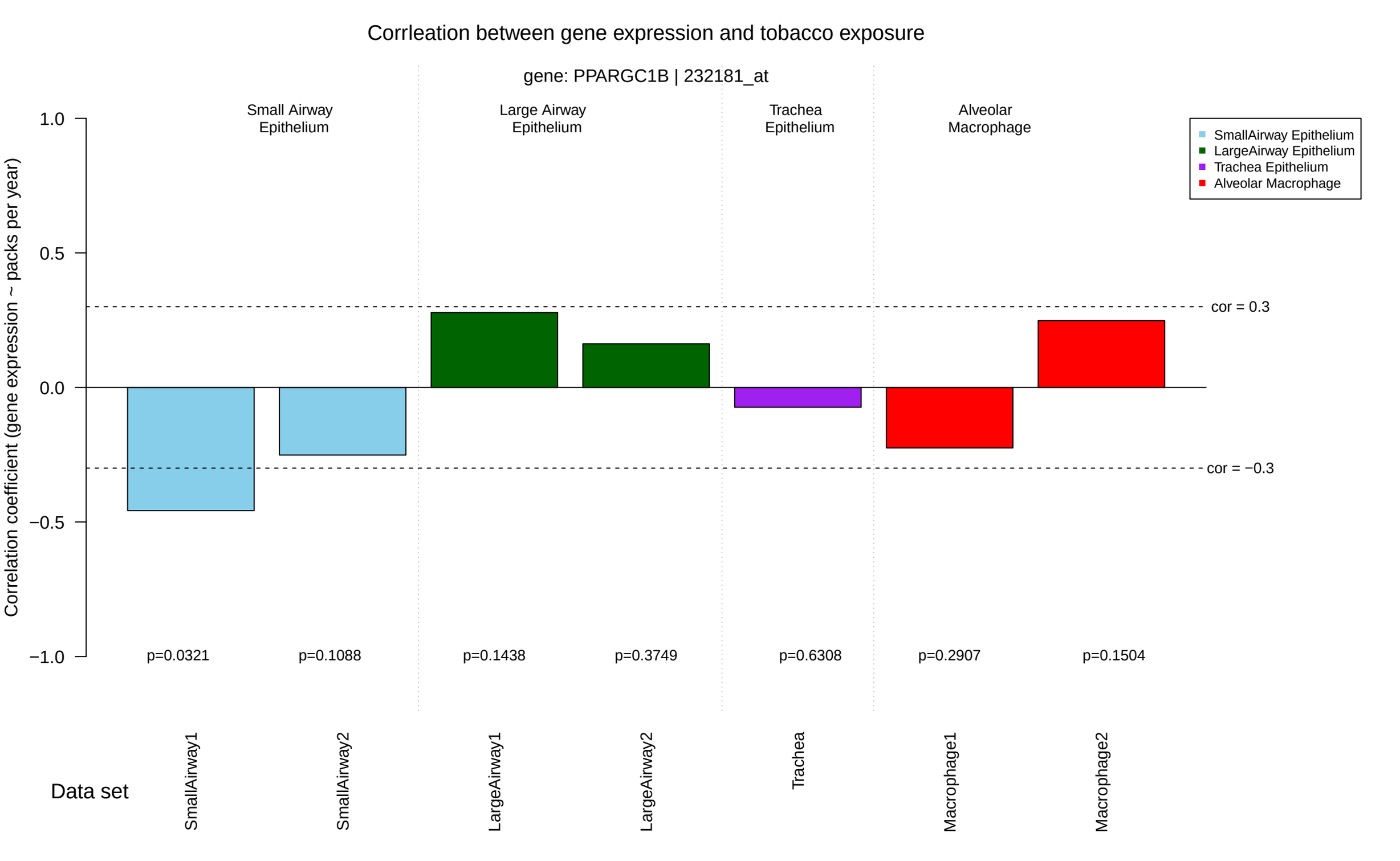

set: LargeAirway1 (p=0.2087)
PPARGC1B | 232181_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9508)
PPARGC1B | 232181_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7247)
PPARGC1B | 232181_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0535)
PPARGC1B | 232181_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1509)
PPARGC1B | 232181_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPARGC1B | 232181_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0321
p=0.1088
p=0.1438
p=0.3749
p=0.6308
p=0.2907
p=0.1504
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2